higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

STRENGTH OF MATERIALS AND STRUCTURES N5

14 AUGUST 2019

This marking guideline consists of 8 pages.

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QUESTION 1

1.1

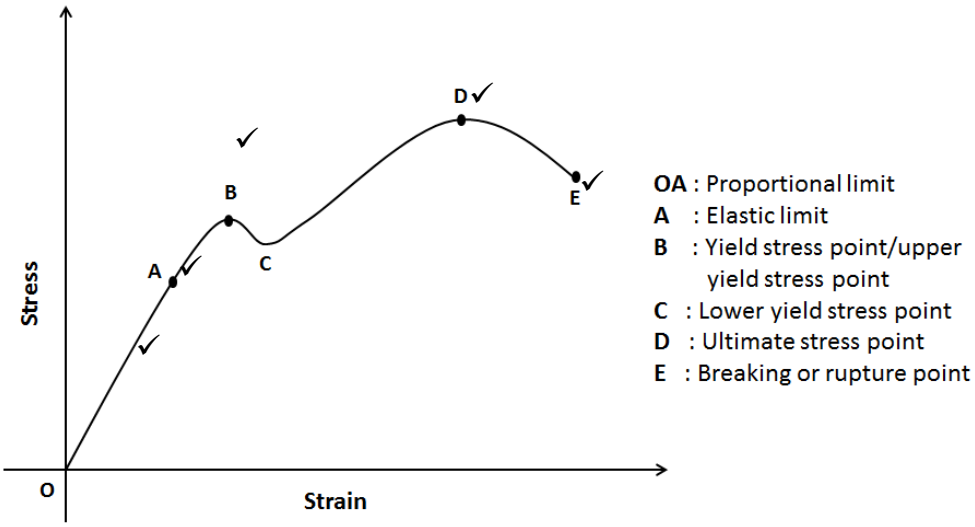


FIGURE 1 (5)

1.2 1.2.1

$$\sigma_{LOP}=\frac{F_{LOP}}{A_O}=\frac{32000}{1,327\times10^{-4}}=241,145\ MPa$$

1.2.2

$$X_{LOP}=\sigma_{LOP}.\left(\frac{L}{E}\right)$$

$$0,22\times10^{-3}=241,145\times10^{6}.\left(\frac{0,07}{E}\right)$$

$$\therefore E=76,728\ GPa$$

1.2.3

$$\sigma_{UTS}=\frac{F_{\max}}{A_O}=\frac{70000}{1,327\times10^{-4}}=527,506\ MPa$$

1.2.4

$$\sigma_F=\frac{F_F}{A_F}=\frac{52000}{8,577\times10^{-5}}=606,077\ MPa$$

1.2.5

$$\%\Delta X=\left(\frac{X}{L}\right).100=\left(\frac{8,33}{13}\right).100=64,077\%$$

(5 × 2) (10)
[15]

QUESTION 2

2.1 2.1.1 $F_t = F_s$

$\sigma_t.(p-d).t = \frac{\pi}{4}.d^2.\tau.n.x$ ✓

$115\times10^6.(p-0,02).0,014 = \left(\frac{\pi}{4}\right).(0,02^2).(75\times10^6).(2).(1,75)$ ✓✓

$\therefore p = 71,222 \approx 71\ mm$ ✓ (4)

2.1.2 Consider plate tension:

$\eta_t = \left(\frac{p-d}{p}\right).100 = \left(\frac{71-20}{71}\right).100 = 71,83\%$ ✓✓✓

Consider shearing:

$\eta_s = \frac{\frac{\pi}{4}.d^2.\tau.n.x}{\sigma_t.t.p} = \frac{\frac{\pi}{4}.(0,02^2).(75\times10^6).(2).(1,75)}{115\times10^6.(0,071).(0,014)} = 72,14\%$ ✓✓

Consider crushing:

$\eta_c = \frac{\sigma_c.d.t.n}{\sigma_t.t.p} = \frac{190\times10^6.(0,02).(0,014).(2)}{115\times10^6.(0,014).(0,071)} = 93,08\%$ ✓✓

Safe joint efficiency $= 71,83\%$ ✓ (10)

[14]

QUESTION 3

3.1 Consider the external load applied: F_t = 240 kN

$$X_1 = X_2 : \frac{F_1 . L_1}{A_1 . E_1} = \frac{F_2 . L_2}{A_2 . E_2} \checkmark$$

$$F_1 = \frac{2,827 \times 10^{-3} . (180 \times 10^9)}{2,262 \times 10^{-3} . (90 \times 10^9)} . F_2 \checkmark \checkmark$$

$$¿ 2,5 . F_2 \overrightarrow{eq1}$$

$$F_t = F_1 + F_2$$

$$240000 = 2,5 . F_2 + F_2 \checkmark$$

$$\therefore F_2 = 68571,429 \ N \checkmark$$

$$\therefore F_1 = 171428,571 \ N \checkmark$$

$$\sigma_2 = \frac{F_2}{A_2} = \frac{68571,429}{2,262 \times 10^{-3}} = 30,315 \ MPa(tensile) \checkmark \checkmark \checkmark$$

Consider the change in temperature: T_1 = 27°C

$$F_1 = F_2 : \sigma_1 . A_1 = \sigma_2 . A_2 \checkmark$$

$$\sigma_1 . (2,827 \times 10^{-3}) = 30,315 \times 10^6 . (2,262 \times 10^{-3}) \checkmark$$

$$\therefore \sigma_1 = 24,256 \ MPa(tensile) \checkmark$$

$$\Delta t . (\alpha_1 - \alpha_2) = \frac{\sigma_1}{E_1} + \frac{\sigma_2}{E_2} \checkmark$$

$$\Delta t . (21 \times 10^{-6} - 16 \times 10^{-6}) = \frac{24,256 \times 10^6}{180 \times 10^9} + \frac{30,315 \times 10^6}{90 \times 10^9} \checkmark \checkmark$$

$$\therefore \Delta t = 94,318^\circ C \checkmark$$

$$t_2 = 27 - 94,318 = -67,318^\circ C \checkmark$$

[19]

QUESTION 4

Configuration diagram

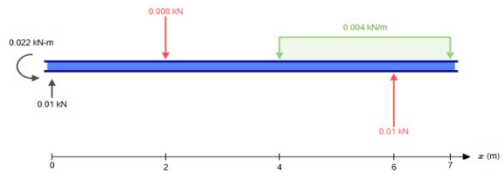


FIGURE 1

4.1

Shear-force Diagram ✓

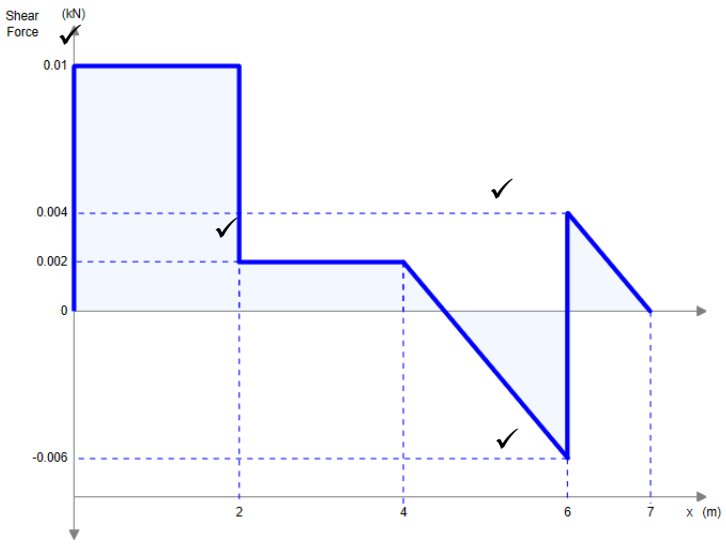


FIGURE 2 (5)

4.2

Bending-moment diagram ✓

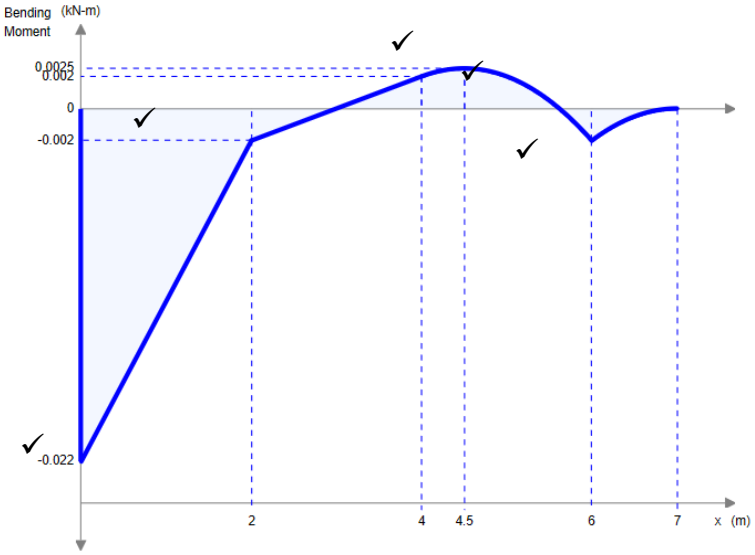


FIGURE 3 (6)

4.3 $\tan\theta = \tan\theta$ ✓

$\frac{2}{x} = \frac{5}{2-x}$ ✓

$2.(2-x) = 5.(x)$

$4 - 2.x = 5.x$

$x = 0,571\ m$ ✓

$\therefore 4,571\ m(\mapsto)$ from fixed support ✓

(4)
[15]

QUESTION 5

5.1 $\overline{y_T} = \frac{\overline{y_1}.A_1 + \overline{y_2}.A_2}{A_1 + A_2} = \frac{734,3.(1,23\times10^{-3}) + 341,75.(17,84\times10^{-3})}{1,23\times10^{-3} + 17,84\times10^{-3}} = 367,069\ mm\uparrow$ ✓✓

$\overline{x_T} = \frac{\overline{x_1}.A_1 + \overline{x_2}.A_2}{A_1 + A_2} = \frac{221,95.(1,23\times10^{-3}) + 126,85.(17,84\times10^{-3})}{1,23\times10^{-3} + 17,84\times10^{-3}} = 132,984\ mm\rightarrow$ ✓✓

(4)

5.2 $I_{XX_T} = \left[I_{X_1} + A_1.r_{y_1}\right] + \left[I_{X_2} + A_2.r_{y_2}\right]$ ✓ ✓ ✓

$= \left[2,176\times10^{-6} + 1,23\times10^{-3}.(0,367^2)\right] + \left[1363\times10^{-6} + 17,84\times10^{-3}.(0,025^2)\right]$ ✓ ✓

$\approx 1,681\times10^{-4} + 1,374\times10^{-3}$ ✓

$\approx 1,543\times10^{-3}\ m^4$

(6)

5.3 $I_{YY_T} = \left[I_{Y_1} + A_1.r_{x_1}\right] + \left[I_{Y_2} + A_2.r_{x_2}\right]$ ✓ ✓

$= \left[0,2528\times10^{-6} + 1,23\times10^{-3}.(0,089^2)\right] + \left[51,83\times10^{-6} + 17,84\times10^{-3}.(0,00613^2)\right]$ ✓ ✓ ✓

$\approx 9,988\times10^{-6} + 5,25\times10^{-5}$

$\approx 6,249\times10^{-5} m^4$ ✓

(6)

5.4 $Z_{max} = \frac{I_{XX_T}}{\overline{y_{max}}} = \frac{1,543\times10^{-3}}{(0,684 + 0,102 - 0,367)} = 3,691\times10^{-3} m^3$ ✓✓

$Z_{min} = \frac{I_{XX_T}}{\overline{y_{min}}} = \frac{1,543\times10^{-3}}{(0,367)} = 4,204\times10^{-3} m^3$ ✓✓

(6)

5.5

$$\sigma_{max}=\frac{M.\overline{y_{max}}}{I_{XX_T}}$$ ✓

$$145\times10^{6}=\frac{M.(0,684+0,102-0,367)}{1,543\times10^{-3}}$$ ✓

$$\therefore M=535211,504\ N.m$$ ✓

$$\omega=(9,65+140)\times9,81=1468,067\ N/m$$ ✓

$$M=\frac{1}{4}.W.L+\frac{1}{8}.\omega.L^{2}$$ ✓

$$535211,504=\frac{1}{4}.(W).(7)+\frac{1}{8}.(1468,067).(7^{2})$$ ✓

$$\therefore W=300,697\ N$$ ✓

(7)
[29]

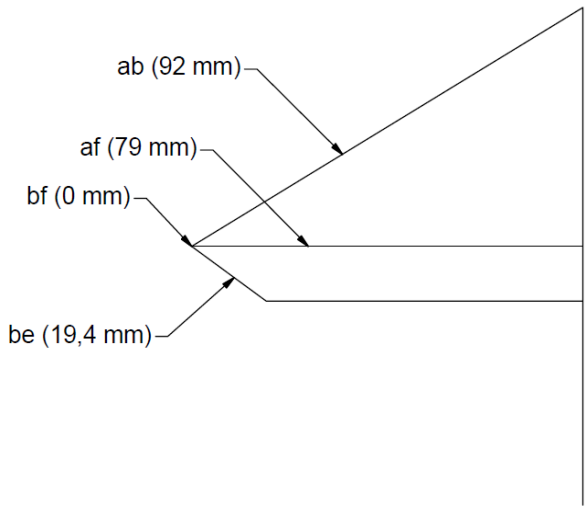
QUESTION 6

6.1 Take moments about LHS support:

$$R_D.(10)=80.(7)+120.(4)$$

$$\therefore R_D=104\ N$$

$$\therefore R_A=96\ N$$



Vector	Magnitude ✓	Nature ✓
ab	184 N ✓	S ✓
af	158 N ✓	T ✓
bf	0 N ✓	✓
be	39 N	S

A tolerance of ±2 N may be used.

Scale 1 mm = 2 N

[8]

TOTAL: 100